MA1513 Weekly Self Practice Set 3

1. Let

$$\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 5 & -1 & 1 \\ 2 & 1 & 8 & -2 & 1 \\ 0 & 1 & 2 & 0 & 1 \\ 1 & 0 & 3 & -1 & 1 \end{pmatrix}.$$

   (i) Use MATLAB or otherwise to find the reduced row echelon form of $\boldsymbol{A}$.

   (ii) Use the answer in (i) to find a basis for the row space of $\boldsymbol{A}$.

   (iii) Find a basis for the column space of $\boldsymbol{A}$.

   (iv) Find a basis for the nullspace of $\boldsymbol{A}$.

2. Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v}_4\}$ be a subset of $\mathbb{R}^6$ where

$$\boldsymbol{v}_1 = (1, 2, -2, 1, 1, 0), \boldsymbol{v}_2 = (-2, -4, 4, -2, -2, 0), \boldsymbol{v}_3 = (2, 1, 1, 2, 0, 1),$$

$$\boldsymbol{v}_4 = (1, -1, 3, 1, -1, 1).$$

   Let $V = \text{span}(S)$, a subspace of $\mathbb{R}^6$.

   (i) Is S a linearly independent set? Why?

   (ii) Find a basis for V that is made up of (some of) the vectors in S. (Refer to section 2.5.)

   (iii) What is the dimension of V?

3. (Use MATLAB with <u>format rat</u> option to find the answers of this question.)

   (i) Let $\boldsymbol{A} = \begin{pmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \\ 40 & 41 \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} 20 \\ 40 \\ 30 \\ 10 \end{pmatrix}$. Find $\boldsymbol{A}^T\boldsymbol{A}$ and $\boldsymbol{A}^T\boldsymbol{b}$.

   (ii) Use (i) to find the least squares solution to the system $\boldsymbol{Ax} = \boldsymbol{b}$.

   (iii) Use (ii) to find the projection of the vector $(20, 40, 30, 10)$ onto the subspace given by $\text{span}\{(10, 20, 30, 40), (11, 21, 31, 41)\}$. (See section 2.7)

   (Give your answers for (ii) and (iii) in <u>fractions</u>)